DeCovarT and CardamomOT: gene selection, noise layers and what a network means

Bastien Chassagnol*

30 October 2026 (draft)

Abstract

DeCovarT (Chassagnol et al. 2023) estimates cell-type fractions from bulk RNA-seq, and models each type by a mean and a gene–gene covariance. CardamomOT [Maugé & Ventre 2026, doi:10.1371/journal.pcbi.1014838] infers a signed, directed gene regulatory network from time-stamped single-cell snapshots, through a mechanistic bursting model. Both pick a small gene panel from a whole transcriptome, and both estimate a gene–gene structure, but for opposite reasons: in DeCovarT the covariance is a nuisance that weights genes, in CardamomOT the network is the estimand. We compare the two along five axes: (i) gene selection, (ii) expression versus measurement model, (iii) a speculative side axis (convolution, Wiener filter, mean-field ansatz), (iv) the diagonal of a network and self-regulation, (v) how many genes the data can support. The CardamomOT code at commit `4a6305c` adds a per-cell depth layer that the paper does not describe, and its diagnostic is the main item DeCovarT could borrow. CardamomOT could borrow a cell-quality tier, a held-out likelihood for its penalty, and a within-basin covariance.

1 Aims, estimands and the two models side by side

Sources. CardamomOT: the published paper (PDF pages cited as 'p.') and the repository at a fixed commit.[1] Gandrillon's sympathoadrenal study [Gandrillon 2025, doi:10.1101/2025.03.21.644507] supplies CardamomOT's gene-selection rule. All PDFs were read with local tools only.

Notation. Genes $g = 1, \dots, G$; cell types $j = 1, \dots, J$ with C_j reference cells; cells c; bulk samples i with n_i cells; reference UMI counts x_{gc} with size factor s_c; fractions $\boldsymbol{p}_{\cdot i}$. DeCovarT's within-type covariance on the linear count scale is $\boldsymbol{\Sigma}_{W,j}$, its precision $\boldsymbol{\Omega}_j = \boldsymbol{\Sigma}_{W,j}^{-1}$, the empirical covariance $\boldsymbol{V}_j$. CardamomOT's interaction matrix is $\boldsymbol{\theta} = (\theta_{gh})$, the influence of gene h on gene g (p. 6). We write the negative binomial as $\mathrm{NB}(a, r)$, shape a and rate r (the paper writes $\mathrm{NB}(a, b)$, the code $\mathrm{NB}(k, c)$), so that the mean is a/r.

The two models answer different questions (Table 1). DeCovarT is a multivariate regression in the strict sense: several correlated responses (the genes) share one design (the fractions). Its gene covariance matters because it weights genes, as in generalised least squares. CardamomOT is a mechanistic model of the cell, and its network is what the user reads. A good covariance for DeCovarT need not be a good network for CardamomOT, and the converse also holds (Section 4).

2 Gene selection (axes i and v)

2.1 CardamomOT: paper and code

The paper treats the panel as prior information. 'All experimental applications presented here operate on gene subsets rather than the full transcriptome (Appendix C), which is not a neutral choice' (p. 27); 'the selected gene set should be recognised as a form of prior information in its own right'

*INSERM. First draft, written with Claude Code (Anthropic). Statements marked [TODO-Vn] wait for the SymPy ledger of Appendix B; none of them is called verified yet.

[1] `https://github.com/eliasventre/CardamomOT`, commit `4a6305cd3699be7a10d8da97b9ad892e41c90adc`. Files read: `estimate_cell_depth.py`, `inference/depth.py`, `inference/depth_model.py`, `inference/gene_selection.py`, `inference/gandrillon.py`, `inference/network.py`. The repository ships no `CITATION.cff`.

Table 1: The two methods side by side. Page numbers refer to the CardamomOT paper unless a file name is given; 'code' means the repository at commit `4a6305c`.

	DeCovarT	CardamomOT
Data	bulk RNA-seq samples and one reference: sorted populations, or the cells of one specimen (Suppinger: single-cell GSE229513, bulk GSE229386, matched at 48 h)	scRNA-seq snapshots at successive time points, raw counts; optional prior network $\boldsymbol{\theta}_0$ and degradation rates (p. 8)
Estimand	fractions $\boldsymbol{p}_{\cdot i}$ of each bulk sample, with Wald intervals	signed directed $\boldsymbol{\theta}$, burst parameters, protein trajectories (pp. 6, 8)
Cell-level law	first two moments $(\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_{W,j})$ per type on the linear UMI scale	$M_g \mid \boldsymbol{P} \sim$ NB from promoter bursting; a mixture over promoter basins, genes independent given the basin (Eq. 4, p. 7)
Aggregate	sum of n_i cells: $\boldsymbol{y}_{\cdot i} \sim \mathcal{N}\big(c_i \sum_j p_{ji}\boldsymbol{\mu}_j,\ \phi_i \sum_j p_{ji}\boldsymbol{\Sigma}_{W,j}\big)$, linear in $\boldsymbol{p}$; population references give $\sum_j p_{ji}^2 \boldsymbol{\Sigma}_j$, quadratic [TODO-V6]	none (no bulk); the population marginal is the basin mixture
Noise layers	expression and measurement mixed in $\boldsymbol{\Sigma}_{W,j}$; bulk scales c_i, ϕ_i	paper: expression (bursting) only; code: per-cell depth factor s_c (Section 3)
Gene selection	QC, presence in every type, consistency screen, optional Fisher-information refinement (Section 2)	paper: core set plus databases (p. 28); code: entropy and Wasserstein terminals, basin separability, coarse network, Steiner closure
Role of the network	nuisance plug-in that weights genes (generalised least squares)	the object of inference
Sparsity	dense shrinkage $\hat{\boldsymbol{\Sigma}}$ by default; graphical lasso with cross-validated λ as an alternative	ℓ_1 towards $\boldsymbol{\theta}_0$ (Eq. 10, p. 9); code: quantile universal threshold per target gene, then an adaptive-lasso refinement (`network.py`)
Diagonal	$\omega_{gg} = 1/\operatorname{Var}(x_g \mid \boldsymbol{x}_{-g})$, never penalised	θ_{gg}, directed self-regulation; 'no specific penalty' (p. 28)
Inference	maximum likelihood, profile likelihood, Wald intervals for $\boldsymbol{p}$	point estimates; run-to-run variability over 3–5 seeds (p. 27)

(p. 27). It recommends a core set of nodes from prior knowledge, completed by curated interaction databases and a coarse genome-wide network, and calls the automation 'ongoing work' (p. 28). For Semrau and Kameneva, the applications use gene lists from earlier work; for Schiebinger, markers plus 'dynamically variable genes' ranked by entropy and Wasserstein distances (p. 32).

The code at `4a6305c` automates that recommendation (`inference/gene_selection.py`, module docstring):

- **gene universe:** mitochondrial, ribosomal-protein and non-coding genes are excluded by name or biotype;
- **terminal genes** from Gandrillon's workflow (`inference/gandrillon.py`): 'KD' is the one-dimensional Wasserstein distance between the counts of a gene at two consecutive time points, divided by the gene's standard deviation over the pair; 'MDE' is the absolute change of a discrete entropy estimated by the BUB estimator [Paninski 2003, Neural Computation, doi to verify]; terminals are then chosen by a round robin over cell types, each tested against the rest by a Wilcoxon test on log-normalised counts (Scanpy `rank_genes_groups`) [Wolf et al. 2018, Scanpy, doi to verify];
- **basin separability:** genes whose negative-binomial modes are well separated (`nb_init_separation`);
- **a coarse network** on highly variable genes (OTVelo-Corr);
- **closure:** a budgeted prize-collecting Steiner selection that anchors the terminals to the stimuli.

Gandrillon's criteria. The 'Kantorovich distance' cited by Gandrillon (p. 7, to Vershik 2006) is the Wasserstein-1 distance, and the code computes it with `scipy.stats.wasserstein_distance`. 'Delta-entropy' is a change of entropy between consecutive time points; the preprint gives no formula, and the code uses the BUB estimate of the Shannon entropy of the counts. Gandrillon's Figure 3 caption reads: 'The union of those two lists gave us a list of 97 genes', the two lists having 200 and 376 genes. A union of these lists has at least 376 genes, so the operation that yields 97 genes is an

intersection. The CardamomOT port states it as such: 'union over transitions of the top_n KD genes, intersected with the union over transitions of the top_n MDE genes'. A second count does not add up either: with five transitions, a union of the top 50 entropy genes per transition has at most 250 genes, not 376. Neither point changes the method; both concern its description. The 97-gene panel contains ten ribosomal-protein genes and *MT-CYB* (Gandrillon, Table 1), which the current universe filter would exclude; the CardamomOT paper notes housekeeping genes such as *RPL30* among the top Kameneva regulators (p. 18).

2.2 DeCovarT

The estimand is a gene panel that makes $(\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_{W,j})$ estimable and informative about $\boldsymbol{p}$, not population-level differential expression. With one specimen, cells are not replicates, so a p-value describes a screen inside that specimen (Zimmerman et al. 2021; Squair et al. 2021). The tiers are written in the feature-selection vignette:

- **QC:** cells by library size, detected genes and mitochondrial fraction; genes by detection;
- **presence in every type**, so that each $\boldsymbol{\Sigma}_{W,j}$ is non-singular; this also removes perfect markers, for which proportions are not identified once gene-wise platform ratios are allowed (Xie et al. 2026, p. 7);
- **consistency screen** ('marker gene consistency', Wang et al. (2019, p. 2)): the between/within ratio on Pearson residuals, or its count-scale versions, the analysis-of-deviance and quasi-likelihood F tests (Lund et al. 2012); the test is a gate, and genes are ranked by effect size;
- **optional refinement** by the Fisher information for $\boldsymbol{p}$, which needs $\boldsymbol{\Sigma}_{W,j}$.

For two types and one gene, under the single-cell model with equal within-type variances σ_g^2, the Fisher information for p is $n_i \Delta_g^2/\sigma_g^2$, with $\Delta_g = \mu_{g1} - \mu_{g2}$ (Appendix A) [TODO-V4]. The ANOVA ratio on C cells per type estimates $(C/2)\Delta_g^2/\sigma_g^2$, so both rank genes alike. With unequal variances or correlated genes they do not: the multivariate information with a common $\boldsymbol{\Sigma}_W$ is $n_i \boldsymbol{\Delta}^\mathsf{T} \boldsymbol{\Sigma}_W^{-1} \boldsymbol{\Delta}$, a Mahalanobis distance. The Hellinger distance between the two type laws is monotone in that distance; the 2-Wasserstein distance is not, since for equal covariances it reduces to $\|\boldsymbol{\Delta}\|$. This links the screen to the overlap criteria of the screening scenario.

The two methods push the within-group variance in opposite directions. CardamomOT keeps genes whose distribution changes in time and splits into separable basins, so multimodality is wanted. DeCovarT wants a unimodal law within each type, with a covariance it can estimate.

2.3 How many genes can the data support?

Three thresholds, from the weakest to the strongest:

- **existence:** the graphical lasso with $\lambda > 0$, positive empirical variances and an unpenalised diagonal has a unique positive-definite solution for any $n \geq 2$ (Carter 2025); convergence says nothing about the graph;
- **maximum-likelihood threshold** $\mathrm{mlt}(G)$: the smallest n for which the unpenalised MLE on a given graph exists almost surely; it equals the largest clique for chordal graphs [Uhler 2012, arXiv:1012.2643; Gross & Sullivant, arXiv:1404.6989]. On graphs selected by the graphical lasso with $p \leq 9$, the MLE never existed at $n = 1$, and the probability that it exists grows with n and with the penalty [Bernstein & Outlaw 2023, arXiv:2312.03145];
- **support recovery:** $n \gtrsim d^2 \log G$, with d the maximum degree, under incoherence and minimum-signal conditions [Ravikumar et al. 2011, arXiv:0811.3628].

CardamomOT has G^2 interaction parameters ('10^4 for $n = 100$', p. 27) and recommends 'not exceeding of the order of 100 genes in the fully unconstrained setting' (p. 28), with more genes allowed when $\boldsymbol{\theta}_0$ fixes more entries. DeCovarT faces the same trade-off per type: several Suppinger types have $C_j < G$, which rules out the unpenalised covariance and puts support recovery out of reach for all but very sparse graphs. Its plug-in only needs a good $\hat{\boldsymbol{\Sigma}}$, not the right graph, which is why dense shrinkage (Schäfer and Strimmer 2005) is the default.

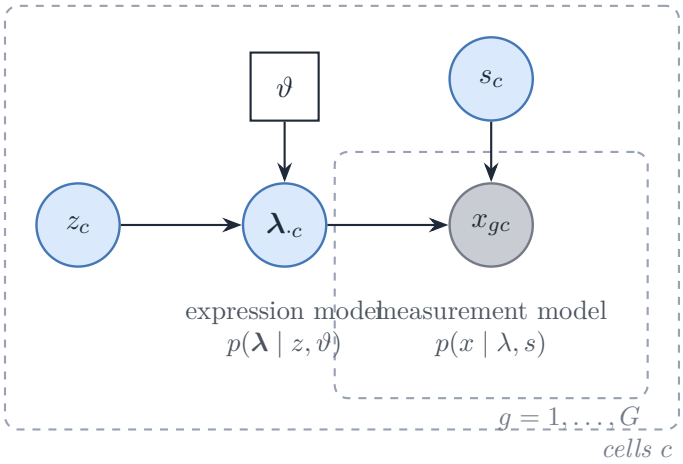


Figure 1: The two noise layers for one reference cell. z_c is the label (cell type j in DeCovarT, promoter basin in CardamomOT), ϑ the parameters ($\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_{W,j}$ in DeCovarT; NB shapes and rates in CardamomOT), s_c the depth factor. CardamomOT's paper models $\boldsymbol{\lambda} \mid z$ as a product of independent NB laws and omits s_c; its code adds s_c. DeCovarT estimates the covariance of x directly, so both layers enter $\boldsymbol{\Sigma}_{W,j}$.

Scalable alternatives. An edge-weighted graphical lasso takes a penalty matrix $\boldsymbol{\Lambda} = (\lambda_{gh})$: small weights for whitelisted pairs, zero constraints for blacklisted ones (Friedman et al. 2008). CardamomOT's prior network, or an OmniPath prior as in its repository (p. 27), can enter DeCovarT this way. Neighbourhood selection fits G lasso regressions [Meinshausen & Bühlmann 2006, doi:10.1214/009053606000000281]; it scales well but returns a graph, not a positive-definite $\hat{\boldsymbol{\Sigma}}$, so a refit under the selected zeros is needed. A penalty chosen for prediction does not select the true neighbourhood (their Proposition 1, p. 5), which matters for CardamomOT, not for DeCovarT's plug-in.

Selection removes what it should estimate. A panel chosen for small within-type variance, such as ReDeconv's smallest coefficients of variation (Lu et al. 2025, p. 14) or MuSiC's 'cross-cell consistent genes' (Wang et al. 2019, p. 3), leaves covariances dominated by Poisson sampling. Their correlations are then attenuated (Section 3), and the estimated precision looks diagonal for reasons that have nothing to do with regulation.

In short: gene selection

- CardamomOT selects for change in time and multimodality, and treats the panel as prior information. DeCovarT selects for between-type separation with an estimable unimodal within-type law. The criteria are not interchangeable.
- The univariate ANOVA ratio is a screen; the Fisher information for $\boldsymbol{p}$ is the objective. They agree for two types only with equal variances and independent genes [TODO-V4].
- Both methods meet the same ceiling: G^2 parameters against C_j cells. Prior structure (a whitelist, $\boldsymbol{\theta}_0$) is the way to more genes.

3 Expression model versus measurement model (axis ii)

Sarkar and Stephens (2021, p. 3) separate 'a measurement model [...] the conditional distribution $p(X \mid \Lambda)$' from 'a model for the expression levels $p(\Lambda)$'. For UMI data, Poisson measurement suffices: $x_{gc} \mid \lambda_{gc} \sim \text{Poisson}(x_{c+}\lambda_{gc})$, and gene–gene correlation belongs to the expression model. Fang and Pachter (2025) add a cell-wide factor $c = c_{\text{tech}} c_{\text{bio}}$ that multiplies the mean of every gene (their eqs 1–2, pp. 3–4). For intrinsically uncorrelated genes, the normalised covariance $\text{Cov}(X_a, X_b)/(\mathbb{E}X_a \mathbb{E}X_b)$ then equals the extrinsic noise $\text{Var}\, c/(\mathbb{E}c)^2$ (eq. 5, p. 5). They estimate it by the mode of the normalised covariances, call 'Poisson genes' those whose bootstrap interval of normalised variance contains it, and take the Poisson MLE of the size factor on those genes (p. 16). Normalising by size factors 'can eliminate biological stochasticity present in extrinsic "noise"' (p. 2).

CardamomOT, paper. The negative binomial comes from bursting and is an expression model

(p. 6). No size factor, library size or capture term appears in Eqs 3–4; technical noise is examined only as a robustness test with simulated binomial dropout of 5–90 % (p. 17).

CardamomOT, code at `4a6305c`. The measurement layer is a per-cell depth factor s_c (`estimate_cell_depth.py`, `inference/depth.py`, `inference/depth_model.py`).

- **Rationale** (script docstring): a depth common to all genes of a cell 'acts like a hidden common regulator: it correlates all genes and can move cells between NB basins'.
- **Estimation** within homogeneous groups (sample, time, cell type), on the whole transcriptome (at least 10,000 genes, otherwise nothing is estimated); the library size leaves out the 50 most expressed genes. The default method, `poissonian`, follows Fang & Pachter: extrinsic noise as the mode of normalised covariances, iterated Poisson genes, s_c as the Poisson MLE on those genes, then a pairwise least-squares rescaling between groups. The alternative, `group_median`, divides the library size by the group median.
- **How it enters:** with the default `depth_model = 'mean'`, counts are $\mathrm{NB}(a, r/s_c)$ (code: $\mathrm{NB}(k, c/s_i)$): depth multiplies the mean and leaves the shape a unchanged, as in scVI. The option `'burst_frequency'` uses $\mathrm{NB}(as_c, r)$. Counts stay integers; s_c is stored in `obs['depth_factor']` and `X` is untouched.
- **When it is applied:** the diagnostic recommends the correction if the depth share of the mean within-group pairwise correlation exceeds 0.5, or if more than half of the genes have NB modes closer than the depth spread. The correlation share is computed on the log1p counts of 1,000 reference genes (the most variable among genes detected in more than 5 % of cells), with and without the log library size regressed out. The depth spread is the median over groups of $q_{0.95}/q_{0.05}$ of s_c. The correction is written only if the user allows it.
- **No cell or gene is removed for abnormal depth.** There is no MAD or quantile filter. Cells flagged as doublets are counted and reported, never filtered; a warning recommends upstream removal when depth correlates with the doublet score above 0.3.

DeCovarT. $\mathbf{\Sigma}_{W,j}$ estimated from raw UMI counts mixes both layers. If $x_{gc} \mid \boldsymbol{\lambda}_{\cdot c}, s_c \sim \mathrm{Poisson}(s_c \lambda_{gc})$ with s_c independent of $\boldsymbol{\lambda}_{\cdot c}$ within type j, the law of total covariance gives (Appendix A)

$$\mathrm{Cov}(\boldsymbol{x}_{\cdot c}) = \mathbb{E}(s^2)\, \mathbf{\Sigma}_j^{\lambda} + \mathrm{Var}(s)\, \boldsymbol{\mu}_j^{\lambda} \boldsymbol{\mu}_j^{\lambda\mathsf{T}} + \mathbb{E}(s)\, \mathrm{diag}(\boldsymbol{\mu}_j^{\lambda}). \tag{1}$$

- **Depth** adds a rank-one term along the mean $\boldsymbol{\mu}_j^{\lambda}$, common to all gene pairs. Its biological part (cell size, the S_j of DeCovarT's estimand) aggregates into the bulk and is wanted. Its technical part (capture) does not reach the bulk. Because the term lies along the type means, an inflated technical share down-weights the very directions that separate the types (our reading, not tested). Splitting the two parts needs controls such as spike-ins or ambient transcripts (Fang and Pachter 2025, pp. 8–11); CardamomOT's diagnostic measures the total depth share only.
- **Poisson sampling** adds to the diagonal only, so it attenuates every correlation and makes lowly expressed panels look independent.
- **Bulk scales** $c_i = a_i n_i$ and $\phi_i = a_i^2 n_i$ absorb a platform scale per bulk sample; they do not remove the cell-level depth variance from $\mathbf{\Sigma}_{W,j}$.
- **A model that separates the layers:** the Poisson log-normal, $x_{gc} \sim \mathrm{Poisson}(s_c e^{z_{gc}})$ with $\boldsymbol{z}_{\cdot c} \sim \mathcal{N}(\boldsymbol{\mu}_z, \mathbf{\Sigma}_z)$ (Chiquet et al. 2021). Its offset is the measurement layer and $\mathbf{\Sigma}_z$ the expression layer. Mapped back to the linear scale, $\mathbb{E}\boldsymbol{x} = s\, e^{\boldsymbol{\mu}_z + \mathrm{diag}(\mathbf{\Sigma}_z)/2}$ and $\mathrm{Cov}\, \boldsymbol{x} = \mathrm{diag}(\mathbb{E}\boldsymbol{x}) + (\mathbb{E}\boldsymbol{x}\, \mathbb{E}\boldsymbol{x}^{\mathsf{T}}) \circ (e^{\mathbf{\Sigma}_z} - 1)$ for fixed s [TODO-V3].

In short: noise layers

- In the paper, CardamomOT's NB is an expression model with no measurement layer. The code at `4a6305c` adds one: a per-cell depth s_c that multiplies the NB mean, applied only when depth explains more than half of the within-group correlation or blurs the NB basins. It removes no cell and no gene; doublets are only flagged.
- DeCovarT's $\mathbf{\Sigma}_{W,j}$ contains a rank-one depth term and a Poisson diagonal. The biological share of depth should stay, the technical share should not.
- DeCovarT can run the depth-share diagnostic on Suppinger before estimating $\mathbf{\Sigma}_{W,j}$. Cell QC stays a separate tier, which CardamomOT does not perform.

4 The diagonal and self-regulation (axis iv)

Two different objects. In a Gaussian graphical model, $\omega_{gg} = 1/\operatorname{Var}(x_g \mid \boldsymbol{x}_{-g})$ is a conditional precision, and an edge is a non-zero partial correlation $-\omega_{gh}/\sqrt{\omega_{gg}\omega_{hh}}$. The diagonal is never an edge. In CardamomOT, θ_{gg} enters the activation rate of gene g through its own protein (Eq. 1, p. 6): a directed self-loop.

Self-activation and bistability. 'The combination of a positive auto-activation with a negative basal parameter β_i is the most parsimonious way to produce bistability for an isolated gene' (p. 28). This is the autocatalytic case of [Ferrell 2002, doi:10.1016/S0955-0674(02)00314-9]: bistability needs feedback, 'some type of non-linearity within the feedback circuit', and balanced legs (p. 3 of that review). CardamomOT reports diagonal coefficients 'systematically non-negligible across all three datasets' (p. 18).

Linking the two through the stationary covariance. Around one basin, linearise the dynamics as an Ornstein–Uhlenbeck process $\mathrm{d}\boldsymbol{x} = \boldsymbol{A}(\boldsymbol{x} - \boldsymbol{x}^*)\mathrm{d}t + \boldsymbol{B}\,\mathrm{d}\boldsymbol{W}$. The stationary covariance solves the Lyapunov equation $\boldsymbol{A\Sigma} + \boldsymbol{\Sigma A}^\mathsf{T} + \boldsymbol{Q} = \boldsymbol{0}$, $\boldsymbol{Q} = \boldsymbol{BB}^\mathsf{T}$. Write $a_g = d_g - \varsigma_g > 0$ for the net decay of gene g: degradation minus linearised self-activation. For two genes with $1 \to 2$ of strength b (Appendix A),

$$\sigma_{11} = \frac{q_1}{2a_1}, \qquad \sigma_{12} = \frac{b\,q_1}{2a_1(a_1+a_2)}, \qquad \omega_{12} = -\frac{2a_2(a_1+a_2)\,b}{(a_1+a_2)^2 q_2 + b^2 q_1} \quad \text{[TODO-V5]}. \tag{2}$$

- Self-activation acts only through a_g: a larger ς_g inflates σ_{gg} and the conditional variance, so it lowers ω_{gg}. It appears nowhere as a 'self-edge'.
- $\omega_{12} \neq 0$ if and only if $b \neq 0$, but its sign and size mix b with the decays and noise intensities, and the direction is lost.
- With three genes, a fork $1 \to 2$, $1 \to 3$ gives $\omega_{23} \propto -b_{21}b_{31} \neq 0$: two co-regulated targets are linked in the precision although neither regulates the other [TODO-V5]. A static structural equation model with the same fork would give $\omega_{23} = 0$.
- The linear model has one basin. Bistability requires the non-linearity that the linearisation removes, and produces a bimodal law within a cell type, which a Gaussian within-type law cannot represent. DeCovarT's counterpart of a basin is a subtype: more components in the convolution.

Penalising the diagonal. When the graphical lasso penalises every entry, its covariance estimate has $\hat{W}_{gg} = V_{gg} + \lambda$ (Friedman et al. 2008): every variance is biased upwards by λ. DeCovarT's likelihood uses those variances as gene weights, so the diagonal is never penalised (`penalize.diagonal = FALSE`). For CardamomOT, the diagonal penalty is a prior choice. The paper says 'we deliberately choose not to apply a specific penalty on diagonal coefficients' (p. 28), and suggests zeroing entries of $\boldsymbol{\theta}_0$ when auto-regulation is implausible. Eq. 10 penalises $\boldsymbol{\theta} - \boldsymbol{\theta}_0$ entrywise. In the code (`network.py`), the first ℓ_1 step penalises the self-loop like any edge inside the prior support, and the adaptive-lasso refinement leaves 'self-regulation and basals unpenalized' (`refine_inference`). Stating that two-stage rule in the paper would remove an ambiguity.

In short: diagonal

- ω_{gg} is a conditional precision; θ_{gg} is a directed self-loop. They are related only through the stationary covariance of a linearised model, where self-activation lowers ω_{gg} [TODO-V5].
- A precision edge is not a regulatory edge: a fork creates one without a direct link.
- Never penalise the GGM diagonal in DeCovarT. In CardamomOT the self-loop penalty is a prior choice, which the code makes differently in its two stages.

5 A side axis (iii): convolution, Wiener filter, mean field

Speculative

These are analogies to explore, not results. Only the Wiener–BLUP identity is a derivation.

- **Convolution and characteristic functions.** The bulk is a sum of independent cell profiles, so its density is a convolution, and its characteristic function is a product: $\varphi_{\boldsymbol{y}}(\boldsymbol{t}) = \prod_j \varphi_{\boldsymbol{x}_j}(p_j \boldsymbol{t})$ for population references, or $\varphi_{\text{cell}}(\boldsymbol{t})^{n_i}$ given a mixture over types for single-cell references (Appendix A). DeCovarT's Gaussian is the central-limit form of that product. For a non-Gaussian cell law, such as a fitted CardamomOT NB mixture, the bulk law of one gene could be computed by FFT on the count lattice [Cooley & Tukey 1965, doi to verify]. In several dimensions the lattice grows exponentially, so this stays a check for a few genes.
- **Wiener filter as BLUP.** Writing $\boldsymbol{x}_{\cdot j,i} = \boldsymbol{\mu}_j + \boldsymbol{\varepsilon}_j$ with $\boldsymbol{\varepsilon}_j \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Sigma}_j)$ makes $\boldsymbol{y} = \boldsymbol{\mu p} + \sum_j p_j \boldsymbol{\varepsilon}_j + \boldsymbol{e}$ a linear mixed model whose random-effect design depends on $\boldsymbol{p}$. Its marginal covariance is $\sum_j p_j^2 \boldsymbol{\Sigma}_j + \boldsymbol{D}$, the best linear unbiased predictor of $\boldsymbol{\varepsilon}_j$ is the Wiener filter of the DeCovarT article, and maximum likelihood is DeCovarT's fit (Appendix A) [TODO-V1]. REML is then an option, and no full Bayesian treatment is needed.
- **Mean field.** Hartree's ansatz writes a many-particle wave function as a product of one-particle functions, each in the average field of the others [Hartree 1928; Fock 1930]. CardamomOT's product over genes given the basin is of that kind: genes interact only through the basin label and the protein field. DeCovarT's independent cells are another product ansatz. In both, the neglected correlation (Fock's exchange term in physics) is the within-basin covariance that CardamomOT reports losing (p. 12).

6 What each method could borrow

DeCovarT from CardamomOT.

- **The depth-share diagnostic**, run on Suppinger per (time, cell type) before $\boldsymbol{\Sigma}_{W,j}$ is estimated. A large share calls for a decision on the technical part of Equation (1), not for blind normalisation, since the biological part reaches the bulk.
- **Prior networks as penalty weights** (OmniPath, as in the CardamomOT repository), entering an edge-weighted graphical lasso.
- **A basin-separability check** within each type: a gene with separated NB modes inside a type signals a subtype that a Gaussian within-type law misrepresents.
- **The panel-as-prior stance** (p. 27): report the panel as part of the model, with the sensitivity of $\hat{\boldsymbol{p}}$ to it.

CardamomOT from DeCovarT.

- **A cell-quality tier** separate from the depth model: the code removes no cell, and Gandrillon's only cell filter was a Scrublet score below 0.2 (his p. 4).
- **A within-basin covariance.** The basin mixture implies a gene covariance that is diagonal plus a between-basin term of rank at most $K - 1$ for K basins, by the law of total covariance; the attenuation of the strongest correlations (p. 12) is the missing within-basin part. A Gaussian copula or a Poisson log-normal layer within each basin would restore it.
- **A held-out likelihood for the penalty**, next to the quantile universal threshold. The expected Gaussian negative log-score $\frac{1}{2}\{\log\det \boldsymbol{\Sigma} + \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\boldsymbol{\Sigma}_{\text{true}})\}$ is minimised at $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}_{\text{true}}$ whatever the law of the cells [TODO-V2], and cross-validation estimates the Kullback–Leibler risk for penalised estimators as well (Commenges and Jacqmin-Gadda 2015); [Commenges et al. 2012, arXiv:1206.1753]

give a fast approximation.

- **An information criterion for the panel**, aimed at the estimand $\boldsymbol{\theta}$ rather than at marginal change in time.
- **Uncertainty for $\boldsymbol{\theta}$** beyond seed-to-seed variability, for instance a bootstrap over cells within time points.

References

Cable, D. M. et al. (2022). 'Robust Decomposition of Cell Type Mixtures in Spatial Transcriptomics'. In: *Nature Biotechnology* 40.4, pp. 517–526. DOI: 10.1038/s41587-021-00830-w.

Cai, B. et al. (2024). 'Statistical Inference of Cell-Type Proportions Estimated from Bulk Expression Data'. In: *Journal of the American Statistical Association* 119.548, pp. 2521–2532. DOI: 10.1080/01621459.2024.2382435.

Carter, J. S. (2025). *Existence of Penalised Likelihood Estimates and Posterior Propriety of Separable Prior Distributions for Gaussian Precision Matrices*. https://arxiv.org/abs/2505.20005v2.

Chassagnol, B., G. Nuel and E. Becht (2023). *DeCovarT, a Multidimensional Probabilistic Model for the Deconvolution of Heterogeneous Transcriptomic Samples*. arXiv. DOI: 10.48550/arxiv.2309.09557.

Chiquet, J., M. Mariadassou and S. Robin (2021). 'The Poisson-Lognormal Model as a Versatile Framework for the Joint Analysis of Species Abundances'. In: *Frontiers in Ecology and Evolution* 9. DOI: 10.3389/fevo.2021.588292.

Chu, T. et al. (2022). 'Cell type and gene expression deconvolution with BayesPrism enables Bayesian integrative analysis across bulk and single-cell RNA sequencing in oncology'. In: *Nature Cancer* 3.4, pp. 505–517. DOI: 10.1038/s43018-022-00356-3.

Commenges, D. and H. Jacqmin-Gadda (2015). *Dynamical Biostatistical Models*. Chapman and Hall/CRC. DOI: 10.1201/b19109.

Dong, M. et al. (2021). 'SCDC: Bulk Gene Expression Deconvolution by Multiple Single-Cell RNA Sequencing References'. In: *Briefings in Bioinformatics* 22. DOI: 10.1093/bib/bbz166.

Erdmann-Pham, D. D. et al. (2021). 'Likelihood-Based Deconvolution of Bulk Gene Expression Data Using Single-Cell References'. In: *Genome Research* 31.10, pp. 1794–1806. DOI: 10.1101/gr.272344.120.

Fang, M. and L. Pachter (2025). *Extrinsic Biological Stochasticity and Technical Noise Normalization of Single-Cell RNA Sequencing Data*. bioRxiv preprint. DOI: 10.1101/2025.05.11.653373.

Friedman, J., T. Hastie and R. Tibshirani (2008). 'Sparse Inverse Covariance Estimation with the Graphical Lasso'. In: *Biostatistics (Oxford, England)* 9. DOI: 10.1093/biostatistics/kxm045.

Guo, S. et al. (2023). 'DeMixSC: A Deconvolution Framework That Uses Single-Cell Sequencing plus a Small Benchmark Dataset for Improved Analysis of Cell-Type Ratios in Complex Tissue Samples'. In: *bioRxiv*, p. 2023.10.10.561733. DOI: 10.1101/2023.10.10.561733.

Lu, S. et al. (2025). 'Transcriptome Size Matters for Single-Cell RNA-seq Normalization and Bulk Deconvolution'. In: *Nature Communications* 16.1. DOI: 10.1038/s41467-025-56623-1.

Lund, S. P. et al. (2012). 'Detecting Differential Expression in RNA-Sequence Data Using Quasi-Likelihood with Shrunken Dispersion Estimates'. In: *Statistical Applications in Genetics and Molecular Biology* 11.5. DOI: 10.1515/1544-6115.1826.

Sarkar, A. and M. Stephens (2021). 'Separating Measurement and Expression Models Clarifies Confusion in Single-Cell RNA Sequencing Analysis'. In: *Nature Genetics* 53.6, pp. 770–777. DOI: 10.1038/s41588-021-00873-4.

Schäfer, J. and K. Strimmer (2005). 'A Shrinkage Approach to Large-Scale Covariance Matrix Estimation and Implications for Functional Genomics'. In: *Statistical Applications in Genetics and Molecular Biology* 4, Article32. DOI: 10.2202/1544-6115.1175.

Squair, J. W. et al. (2021). 'Confronting False Discoveries in Single-Cell Differential Expression'. In: *Nature Communications* 12.1. DOI: 10.1038/s41467-021-25960-2.

Tsoucas, D. et al. (2019). 'Accurate Estimation of Cell-Type Composition from Gene Expression Data'. In: *Nature Communications* 10. DOI: 10.1038/s41467-019-10802-z.

Wang, X. et al. (2019). 'Bulk Tissue Cell Type Deconvolution with Multi-Subject Single-Cell Expression Reference'. In: *Nature Communications* 10. DOI: 10.1038/s41467-018-08023-x.

Xie, D., L. Gui and J. Wang (2026). 'Statistical Inference for Cell Type Deconvolution'. In: *Journal of the Royal Statistical Society Series B: Statistical Methodology* 88.4, pp. 1449–1468. DOI: 10.1093/jrsssb/qkag054.

Zimmerman, K. D., M. A. Espeland and C. D. Langefeld (2021). 'A Practical Solution to Pseudoreplication Bias in Single-Cell Studies'. In: *Nature Communications* 12.1, p. 738. DOI: 10.1038/s41467-021-21038-1.

A Derivations

A.1 Two-gene Lyapunov equation. With $\boldsymbol{A} = \left(\begin{smallmatrix} -a_1 & 0 \\ b & -a_2 \end{smallmatrix}\right)$ and $\boldsymbol{Q} = \operatorname{diag}(q_1, q_2)$, the three distinct entries of $\boldsymbol{A\Sigma}+\boldsymbol{\Sigma A}^\mathsf{T}+\boldsymbol{Q} = \boldsymbol{0}$ read $-2a_1\sigma_{11}+q_1 = 0$, $b\sigma_{11}-(a_1+a_2)\sigma_{12} = 0$ and $2b\sigma_{12}-2a_2\sigma_{22}+q_2 = 0$. Hence σ_{11} and σ_{12} as in Equation (2) and $\sigma_{22} = \{a_1(a_1+a_2)q_2 + b^2q_1\}/\{2a_1a_2(a_1+a_2)\}$. Inverting gives

$$\omega_{11} = \frac{2\{a_1(a_1+a_2)^2q_2 + (a_1+a_2)b^2q_1\}}{q_1\{(a_1+a_2)^2q_2 + b^2q_1\}}, \qquad \omega_{22} = \frac{2a_2(a_1+a_2)^2}{(a_1+a_2)^2q_2 + b^2q_1},$$

and ω_{12} of Equation (2). For the fork $1 \to 2$ (strength b), $1 \to 3$ (strength c), $\omega_{23} = -4a_2a_3bc\,q_1(a_1 + a_2)(a_1+a_3)(a_2+a_3)/\mathcal{D}$ with $\mathcal{D} > 0$ a polynomial in the rates and noise intensities [TODO-V5]. These forms come from an exploratory SymPy run, not yet from the ledger.

A.2 Gaussian convolution. Population references: $\boldsymbol{y} = \sum_j p_j\boldsymbol{x}_j$ with independent $\boldsymbol{x}_j \sim \mathcal{N}(\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)$ gives $\varphi_{\boldsymbol{y}}(\boldsymbol{t}) = \prod_j \varphi_{\boldsymbol{x}_j}(p_j\boldsymbol{t}) = \exp\{i\boldsymbol{t}^\mathsf{T}\sum_j p_j\boldsymbol{\mu}_j - \frac{1}{2}\boldsymbol{t}^\mathsf{T}(\sum_j p_j^2\boldsymbol{\Sigma}_j)\boldsymbol{t}\}$, hence the quadratic covariance. Single-cell references: given m_j cells of type j, the sum has mean $\sum_j m_j\boldsymbol{\mu}_j$ and covariance $\sum_j m_j\boldsymbol{\Sigma}_{W,j}$; with $m_j \approx n_ip_j$ and the platform scale a_i, the covariance is $\phi_i\sum_j p_j\boldsymbol{\Sigma}_{W,j}$, linear in $\boldsymbol{p}$ [TODO-V6].

A.3 Wiener filter as BLUP. Stack $\boldsymbol{u} = (\boldsymbol{\varepsilon}_1, \ldots, \boldsymbol{\varepsilon}_J)$ with $\boldsymbol{G} = \operatorname{blockdiag}(\boldsymbol{\Sigma}_j)$ and $\boldsymbol{Z} = (p_1\boldsymbol{I}_G, \ldots, p_J\boldsymbol{I}_G)$. Then $\boldsymbol{y} = \boldsymbol{\mu p}+\boldsymbol{Zu}+\boldsymbol{e}$, $\boldsymbol{e} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{D})$, and $\boldsymbol{V} = \operatorname{Var}\boldsymbol{y} = \boldsymbol{ZGZ}^\mathsf{T}+\boldsymbol{D} = \sum_j p_j^2\boldsymbol{\Sigma}_j+\boldsymbol{D}$. The BLUP is $\hat{\boldsymbol{u}} = \boldsymbol{GZ}^\mathsf{T}\boldsymbol{V}^{-1}(\boldsymbol{y} - \boldsymbol{\mu p})$, whose block j is $p_j\boldsymbol{\Sigma}_j\boldsymbol{V}^{-1}(\boldsymbol{y} - \boldsymbol{\mu p})$: the Wiener filter of the DeCovarT article (§ Wiener filter), equal to the posterior mean in the Gaussian model [TODO-V1].

A.4 ANOVA ratio and Fisher information, $J = 2$. One gene, single-cell model: $y \sim \mathcal{N}(c\{p\mu_1 + (1-p)\mu_2\}, \phi\{p\sigma_1^2 + (1-p)\sigma_2^2\})$. For $\mathcal{N}(m(p), v(p))$, $I(p) = m'(p)^2/v + \frac{1}{2}\{v'(p)/v\}^2$, so $I(p) = c^2\Delta^2/v + \frac{1}{2}\{(\sigma_1^2-\sigma_2^2)/(p\sigma_1^2 + (1-p)\sigma_2^2)\}^2$. If $\sigma_1^2 = \sigma_2^2 = \sigma^2$, the second term vanishes and $I(p) = (c^2/\phi)\Delta^2/\sigma^2 = n_i\Delta^2/\sigma^2$, since $c^2/\phi = n_i$. With C reference cells per type, the ANOVA ratio estimates $(C/2)\Delta^2/\sigma^2$, so $I(p) \approx (2n_i/C)\,F$. Over independent genes the informations add; with correlated genes they do not [TODO-V4]. Under the population model, $v(p) = p^2\sigma_1^2 + (1-p)^2\sigma_2^2$ depends on p even when $\sigma_1 = \sigma_2$, and the proportionality fails.

A.5 Depth and sampling in the within-type covariance. Conditionally on $(\boldsymbol{\lambda}, s)$, counts are independent Poisson with means $s\boldsymbol{\lambda}$. Then $\mathbb{E}\boldsymbol{x} = \mathbb{E}(s)\boldsymbol{\mu}^\lambda$ and $\operatorname{Cov}\boldsymbol{x} = \mathbb{E}\{\operatorname{Cov}(\boldsymbol{x} \mid \boldsymbol{\lambda}, s)\} + \operatorname{Cov}\{\mathbb{E}(\boldsymbol{x} \mid \boldsymbol{\lambda}, s)\} = \mathbb{E}(s)\operatorname{diag}(\boldsymbol{\mu}^\lambda) + \operatorname{Cov}(s\boldsymbol{\lambda})$, with $\operatorname{Cov}(s\boldsymbol{\lambda}) = \mathbb{E}(s^2)\boldsymbol{\Sigma}^\lambda + \operatorname{Var}(s)\boldsymbol{\mu}^\lambda\boldsymbol{\mu}^{\lambda\mathsf{T}}$ when s and $\boldsymbol{\lambda}$ are independent; this is Equation (1). Dividing the off-diagonal terms by the means recovers Fang & Pachter's normalised covariance $\operatorname{Var}(s)/(\mathbb{E}s)^2$ for genes with $\Sigma^\lambda_{gh} = 0$, up to the normalisation $\mathbb{E}s = 1$.

B Verification ledger

The SymPy checks V1–V6 are written last, after every other task of the plan. Until then each statement that depends on one carries a [TODO-Vn] marker. V1: LMM covariance and BLUP = Wiener filter; V2: Stein's loss minimised at the true covariance; V3: Poisson log-normal moment map; V4: Fisher information against the ANOVA ratio for $J = 2$; V5: two-gene Lyapunov covariance and precision, fork precision; V6: single-cell linear against population quadratic covariance at $J = 2$.

Ledger pending.

C Gene selection in second-generation deconvolution methods

Table 2 summarises the review behind Section 2. Pages are PDF pages of the local files (main text unless 'S' for the supplement); a dash means not reported in the parts we extracted.

Table 2: Gene and cell selection in deconvolution methods, and CardamomOT for comparison. QC: quality control; WLS: weighted least squares; GLS: generalised least squares.

Method	Cell QC	Gene filter and marker rule	Cross-platform	Weighting	Gene correlation
RNA-Sieve (Erdmann-Pham et al. 2021)	optional: library size ≥ 3 MAD from the median (S p. 3)	$\geq 20\,\%$ non-zero in one type (S p. 3); bulk outside a sub-Gaussian range, MAD outliers on four statistics (eqs 12–13, p. 11)	range bound on the bulk	asymptotic Gaussian composite likelihood	ignored
DECALS (Cai et al. 2024)	–	CIBERSORTx S-mode genes $\cap$ Fisher-exact markers (p. 13; S pp. 18–19)	–	constrained OLS/GLS	type covariances, least squares and SCAD thresholding (p. 9)
MEAD (Xie et al. 2026)	–	–; 'not identified if all genes are perfect marker genes' (p. 7)	gene-specific platform ratios with mean 1 (pp. 5–6)	diagonal weights, sandwich	dependence set by multiple testing
ReDeconv (Lu et al. 2025)	–	t-test $p < 0.05$ against every other type, no FDR; fold change ≥ 2; smallest CV; about 100 genes per type (p. 14)	TPM/FPKM on bulk only (p. 2)	Gaussian (CLT), heteroscedastic	independent genes
MuSiC (Wang et al. 2019)	–	all genes by default (p. 7); low within-cluster variance in tree-guided stages (pp. 3, 8)	gene- and subject-specific intercept (p. 7)	W-NNLS, cross-subject variance (Eq. 10, p. 7)	diagonal
deMixSC (Guo et al. 2023)	< 500 UMI, < 200 genes, $> 50\,\%$ mitochondrial (p. 11)	bulk vs pseudo-bulk paired t-test, BH (pp. 4, 12)	discrepant genes down-weighted by $a = 1000$ (p. 12)	WLS	diagonal
BayesPrism (Chu et al. 2022)	–	ribosomal, mitochondrial (pp. 14, 19) and chrX/Y genes removed (p. 14); optional scran markers (p. 14)	weak fold-change prior	Bayesian multinomial	sum constraint only (our reading)
DWLS (Tsoucas et al. 2019)	–	MAST hurdle FDR < 0.01, log2 FC > 0.5; smallest condition number (p. 7)	–	WLS, dampening constant by cross-validation (p. 7)	diagonal
SCDC (Dong et al. 2021)	–	–	several references	maximal-variance weights, clipped at the 15th/85th percentiles (p. 10)	diagonal
RCTD (Cable et al. 2022)	–	–	gene-wise platform random effect (pp. 3, 6)	Poisson–lognormal GLMM	–
CardamomOT (code)	none removed; doublets flagged	mitochondrial, ribosomal, non-coding excluded; W1 and entropy terminals; Wilcoxon round robin; NB separability; Steiner closure	not applicable (no bulk)	NB-mixture likelihood	none within a basin (p. 12)